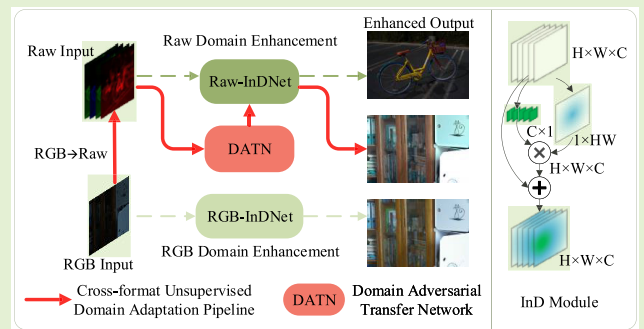


Bridging the Gap Between Low-Light Raw and RGB Image Enhancement Using Domain Adversarial Transfer Network

Pengliang Tang^{ID}, Jiangbo Pei^{ID}, Jianan Han^{ID}, and Aidong Men^{ID}

Abstract—Images captured under low-light conditions often suffer from a low signal-to-noise ratio (SNR) caused by low photon count, making the low-light image enhancement (LLIE) task very challenging. In this work, we contribute to the LLIE task in two dimensions. First, we propose a plug-and-play information integration-and-diffusion (InD) module to address the detail and color reconstruction problems of existing methods for supervised single-format LLIE (i.e., Raw or RGB format). The InD module uses carefully designed matrix multiplications to efficiently extract features that integrate global and pixel-level information. On top of this, we build a novel cross-format unsupervised domain adaptation (CUDA) framework to bridge the domain gap and tackle the unsupervised RGB format LLIE task by fully leveraging the Raw priors inherent in the pretrained Raw domain networks. Specifically, in the first stage, we train an RGB-to-Raw format conversion network to eliminate the format differences. Then, an unsupervised domain adversarial transfer network (DATN) is employed to decrease the feature distance between the target domain (RGB domain) data and the source domain (Raw domain) data. At last, the domain transferred low-light images are enhanced by the pretrained source domain network. Comprehensive experimental results show that the networks equipped with our InD modules outperform state-of-the-art supervised LLIE approaches on both RGB and Raw datasets. Moreover, our CUDA framework also achieves state-of-the-art unsupervised results on RGB datasets.

Index Terms—Adversarial learning, cross-format unsupervised domain adaptation (CUDA), global information extraction, low-light image enhancement (LLIE).



I. INTRODUCTION

WHILE the demand for low-light imaging is extensive, such as for industry, military, and entertainment use, the task itself is extremely challenging due to the low signal-to-noise ratio (SNR) caused by low photon count. Physical means to improve SNR have their characteristic drawbacks, e.g., shooting with high ISO amplifies noise, and extending the exposure time requires the targets to remain absolutely still. Hence, low-light image enhancement (LLIE) methods are necessitated as an alternative for recovering images taken in the low-light environment.

In the past decades, numerous methods have been proposed to address the LLIE problem. The recent comprehensive

development of deep learning-based methods has broken the dominance of classic traditional methods, such as methods based on histogram equalization (HE) [1], [2] and Retinex theory [3], [4]. To improve the performance of RGB format LLIE, a series of methods combine the designs of deep networks with the Retinex theory, e.g., [5], [6], [7]. Also, many state-of-the-art deep learning-based technologies have been introduced, such as transformer [8] and normalizing flow [9]. On the other hand, Chen et al.'s study [10] discovers that the Raw sensor data input produces significantly superior LLIE results compared with the RGB data input. Later research [11] elaborates that in LLIE scenarios, Raw sensor data possesses superiorities over RGB format data due to the availability of shooting parameters in the meta-data, the linear nature of the sensor data, and a fine-grained quantization level. Following Chen et al. [10], a variety of works such as [12], [13], [14], [15], [16], apply end-to-end deep neural networks to reconstruct high-quality images from Raw sensor data.

Although these methods have achieved significant progress, two major challenges still exist as follows.

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The authors are with the School of Artificial Intelligence, Beijing University of Posts and Telecommunications, Beijing 100876, China (e-mail: tangpengliang@bupt.edu.cn; menad@bupt.edu.cn).

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- 1) For supervised single-format LLIE, whether with Raw or RGB inputs, most existing methods exhibit potential for further improvement in the restoration of detailed textures and colors.
- 2) Although it has been proved that the Raw format outperforms the RGB format in LLIE tasks, it is worth emphasizing that RGB format images dominate in practical applications. Therefore, how to bridge the domain gap and introducing the advantages of Raw domain priors to user-friendly RGB format to improve its LLIE performance has become a new challenge.

For the first challenge, due to the low SNR of images taken under extremely low illumination conditions, we commonly observe the loss of details, color distortion, and poor color smoothness in the results of existing methods. In order to address these issues, some existing approaches introduce global information extraction modules in the LLIE network design. Zhang et al. [16] combines a nonlocal (NL) operation [17] and a channel attention mechanism to settle color artifacts, but the vast computation cost introduced by the NL operation remains to be attended. Meng et al. [14] introduced a simple global information extraction method similar to the simplified NL (SNL) [18] to improve the overall performance, but still failed at recovering accurate details and smoothing colors. By observing their experimental results and analyzing their network structures, we attribute their shortcomings to the lack of efficient feature extraction that considers both global and detailed information. Based on the above analysis, in this article, we propose a pixel-intensive global information extraction module for LLIE networks to recover preeminent image details and colors, namely, the information integration-and-diffusion (InD) module. Aiming at calculating a global information value for each pixel while avoiding the huge computational cost brought by calculating intense pixel-to-pixel relations, the proposed InD module integrates the spatial-wise and channel-wise information separately and then diffuses them to each other by a matrix multiplication operation, resulting in a pixel-intensive global information matrix with the same shape as the input feature maps.

For the second challenge, we notice the respective advantages of Raw format and RGB format in the LLIE task. However, the domain gap between them limits the cross-format generalization of LLIE methods, resulting in their advantages being unable to complement for each other. In addition, due to the considerable effort required to construct low and normal-light paired datasets, the unsupervised RGB format LLIE holds even greater practical value. Therefore, in this article, we propose a novel RGB-to-Raw cross-format unsupervised domain adaptation (CUDA) framework to tackle the domain gap and introduce Raw domain priors into the unsupervised RGB format LLIE task. First, we train an RGB-to-Raw image format conversion network (IFCN) on the Raw domain dataset to tackle the format disparity. Then, we design an unsupervised domain adversarial transfer network (DATN) to address the cross-format domain differences, which include differences in dataset scenarios, shooting illumination, pre-processing operations, etc. Finally, a pretrained Raw domain

InDNet, denoted as Raw-InDNet, is implemented to enhance the low-light domain-transferred images of the RGB domain. Thus far, we successfully bridged the domain gap and the Raw prior has been utilized to improve the performance of the unsupervised RGB LLIE in the form of pretrained Raw domain LLIE network.

This work is the extension of our previous conference paper [19]. Compared to the conference version, this work further makes significant contributions. First, the original InDNet is designed for Raw datasets only. However, additional experiments demonstrate that, with minor adjustments, the proposed InDNet can also achieve state-of-the-art results on RGB datasets. More importantly, we design a brand new cross-format domain adaptation framework to bridge the domain gap between the Raw and RGB format LLIE tasks, making our proposed InDNet trained on Raw sensor data successfully generalize to RGB data and improving the unsupervised performance of RGB LLIE.

To the best of our knowledge, this is the first attempt to tackle the cross-format LLIE generalization problem by using the unsupervised domain adaptation (UDA) techniques.

In summary, the main contributions of our work are as follows.

- 1) We propose an InD module to gather pixel-intensive global information to address the detail and color reconstruction issues of Raw and RGB LLIE tasks.
- 2) We design a novel RGB-to-Raw CUDA framework with a DATN to bridge the domain gap between Raw and RGB images and improve the performance of unsupervised RGB format LLIE by introducing Raw priors inherent in the pretrained Raw domain LLIE networks.
- 3) Experimental results demonstrate that LLIE networks equipped with InD modules acquire great performance improvement and outperform state-of-the-art approaches when trained with paired images. Moreover, our CUDA framework also achieves state-of-the-art unsupervised results on RGB datasets.

II. RELATED WORKS

A. LLIE Methods for RGB Images

As the most common form of image presentation, RGB images have long been the main research object of LLIE tasks. Typical traditional techniques focus on manipulating the global brightness and contrast of RGB images. Among them, the most classic methods are HE-based methods and Retinex-based methods. HE-based LLIE methods [1], [2], [20] use the cumulative distribution function to adjust the output gray levels to enhance hidden details in dark areas. Retinex-based LLIE methods are based on the Retinex theory of color vision [3], the essence of which is to determine the reflection map of an image by removing the effects of the low-light illumination map from the image. Lee et al. [21] first introduced the Retinex theory into image enhancement tasks. A series of subsequent studies design more refined structures and hand-crafted priors to achieve satisfactory LLIE results, e.g., multiscale Retinex [22], robust Retinex [23], and structure-aware prior [24]. These methods often assume low-light images are noise-free or the noise is removed by an

additional denoising process. Thus, they fail to recover low SNR images captured under limited illumination.

In the circumstances of the comprehensive development of deep neural networks, a series of deep learning-based LLIE models [5], [6], [7], [8], [9], [25], [26] have been developed since LLNet [27]. They rely on sophisticated designed networks, learning strategies, and most importantly, the learned mapping of the paired low and normal-light images. Some of them combine the designs of deep networks with the Retinex theory for better illumination adjustment and degradation recovery, e.g., RetinexNet [5], KinD [6], and KinD++ [7]. Also, many state-of-the-art deep learning-based technologies have been introduced for LLIE tasks, such as knowledge distillation [28], transformer [8], and normalizing flow [9]. In addition to supervised ones, methods based on unsupervised learning strategies such as [29], [30], [31], [32], [33], have been developed to overcome the challenge of lacking paired low and normal-light data. EnlightenGAN [30] proposes a generative adversarial network that regularizes unpaired learning using a multiscale discriminator, RUAS [31] develops a Retinex-inspired unrolling model with neural architecture search for unsupervised LLIE task.

Compared with the method described above, we design a more refined feature extraction module, which can efficiently extract features that integrate global and pixel-level information to address the existing detail and color restoration problem for the supervised LLIE task. Also, we set the precedent of utilizing the complementary strengths of Raw and RGB formats in the LLIE task, so that the highly informative Raw prior can be introduced into the user-friendly RGB format LLIE task. As a result, the proposed CUDA framework successfully generalizes the well-trained Raw format LLIE network to RGB and improves the performance of the unsupervised RGB format LLIE task.

B. LLIE Methods for Raw Images

Recently, Chen et al. [10] proposed to reconstruct high-quality images from extremely low-light Raw sensor data. This pioneering work opens a new era for LLIE tasks as Raw sensor data covers a wider color gamut and higher dynamic range than RGB data, hence deep models trained on Raw data usually recover better details, colors, and contrast [34]. Following Chen et al. [10], a variety of works apply end-to-end deep neural networks to handle Raw format LLIE tasks, e.g., EEMEFN [12] combines multiexposure fusion and edge detection for better local illumination and edge enhancement, LDC [13] proposes a frequency-based decomposition-and-enhancement network to suppress noise and restore details at different frequencies, GIA-Net [14] introduces global information to solve the color inconsistency problem and ALEN [16] combines NL operation with a channel attention mechanism to settle color artifacts. Most recent work [11] elaborates on the superiority of using Raw images for the LLIE task quantitatively and proposes to utilize Raw files to assist in training the RGB-based LLIE method.

Compared with GIA-Net and ALEN, the proposed InD module is more efficient in obtaining global information and innovatively allocating the global information to each pixel.

In addition, different from [11], which also includes Raw information for RGB format LLIE task, we aim to address the unsupervised RGB format LLIE task by leveraging the Raw priors inherent in the pretrained Raw domain LLIE network, rather than directly using the meta-data of Raw images for supervised training. In order to solve the format and distribution differences between Raw and RGB inputs in the LLIE task, we innovatively define the prior introduction from Raw to RGB as a domain adaptation problem. To the best of our knowledge, this is the first attempt to utilize UDA techniques to deal with the unsupervised cross-format generalization problem of the LLIE task.

C. Domain Adaptation Methods

Domain adaptation generally seeks to learn a model from a source labeled data that can be generalized to a target domain by minimizing the difference between domain distributions [35]. Based on whether to use neural networks as domain alignment techniques, existing approaches can be roughly divided into shallow and deep domain adaptation approaches. The mostly used distance measures in shallow domain adaptation to align the distributions include maximum mean discrepancy [36], Wasserstein metric, correlation alignment [37], Kullback–Leibler divergence [38], and contrastive domain discrepancy [39]. As for deep domain adaptation approaches, DAN [40] utilizes a convolutional neural network, Glorot et al. [41] utilizes an auto-encoder, and DANN [42] utilizes an adversarial based network to diminish the domain gap. Recently, a series of domain adaptation methods for pixel-level tasks have been proposed. PixelDA [43] proposes an unsupervised pixel-level domain adaptation method that utilizes generative adversarial networks to adopt one domain to another. The domain transfer network [44] is a feature-level generative-based adaptation that enforces consistency in the embedding space to transform a source image into a target image. In the LLIE field, Prabhakar et al. [45] recently proposed a few-shot domain adaptation method for cross-camera Raw LLIE, and Jin et al. [46] provided a hyperparameter optimization perspective to formulate a bilevel learning scheme for fast adaptation of general low-light scenarios.

Different from [45] and [46], they perform supervised domain adaptation within the same format, one is for different Raw formats, and the other is for RGB images of different scenes, we perform cross-format UDA for unsupervised LLIE task. The innovation of our use of UDA technology lies in the following two points. First, instead of directly using the existing UDA framework, we designed a brand new IFCN for cross-format conversion requirements and proposed several customized structures in the design of DATN for better domain transfer in the LLIE task. In addition, we also modify the optimization strategy of CUDA to obtain better-unsupervised training results with smaller training costs. More details will be elaborated in Section IV.

III. IND NETWORK (INDNET) FOR SINGLE FORMAT LLIE

In this section, we first introduce the overview pipeline for single-format supervised LLIE tasks. Subsequently, we present

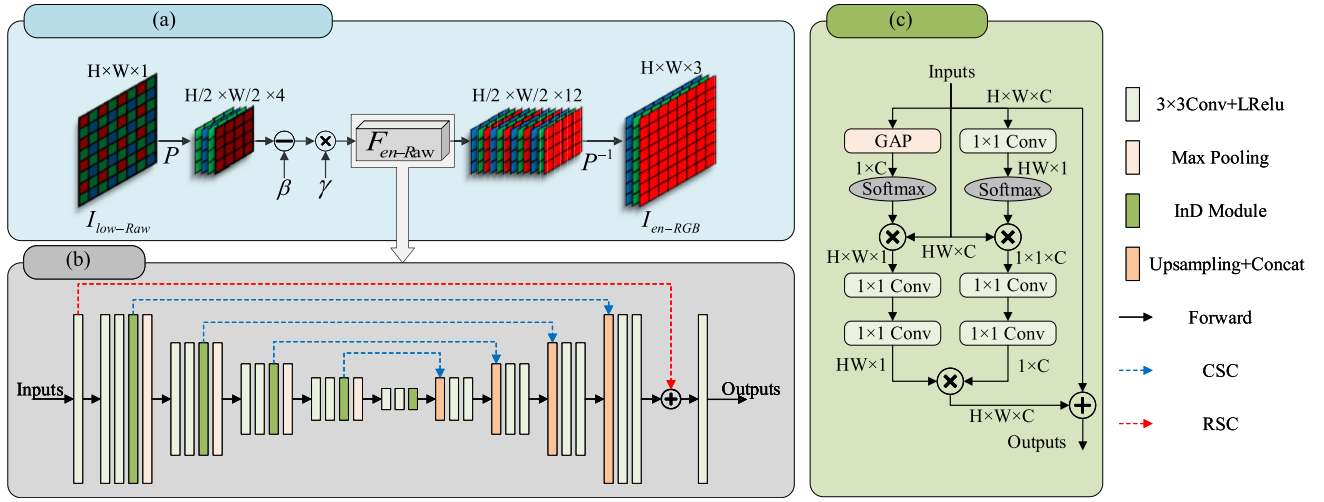


Fig. 1. Illustration of the Bayer Raw format LLIE pipeline in (a), the proposed InDNet in (b), and the detailed InD module in (c). In (a), P denotes the packing operation and P^{-1} denotes the corresponding unpacking operation. β denotes the black level of the shooting device, and γ denotes the exposure time ratio. Both of them are provided by the Raw dataset in this pipeline. F_{en-Raw} refers to the neural network for Raw LLIE, which is InDNet in this article. In (b), multiple plug-and-play InD modules are inserted at different stages of the baseline U-net, forming the InDNet. The InDNet is trained supervisely by $\mathcal{L}_{en}$ in (4). In (c), $\otimes$ represents matrix multiplication and $\oplus$ denotes element-wise addition. The abbreviation GAP stands for global average pooling, CSC stands for concatenate skip connection, and RSC stands for residual skip connection.

the architecture of the proposed InDNet. It is important to note that the well-trained InDNet can function as a standalone LLIE network for single-format data, namely, Raw or RGB data. In addition, with fixed parameters, it can also serve as the predictor part of our proposed CUDA framework for unsupervised cross-format LLIE. Then, we present the detailed structure of the proposed InD module. Finally, we describe the loss function employed to train the proposed InDNet.

A. Pipelines for Single Format LLIE

In this article, we follow the image processing pipeline outlined in [10] to address the extreme LLIE task on Raw sensor data. The pipeline is depicted in Fig. 1(a). Given a low-light one-channel Raw image $I_{low-Raw}$, the three-channel enhanced RGB image I_{en-RGB} can be defined as

$$I_{en-RGB} = P^{-1} (F_{en-Raw} (\gamma \times (P(I_{low-Raw}) - \beta))) \quad (1)$$

where $F_{en-Raw}(\cdot)$ refers to the LLIE network for Raw format. P represents the packing operation, which, taking the Bayer Raw as an example, reshapes the one-channel Raw inputs into four channels and correspondingly reduces the spatial resolution by a factor of two. The packed inputs are first subtracted by the black level β , then scaled by the exposure time ratio γ (e.g., $\times 100$ or $\times 300$). Please note that these parameters, β , and γ , are provided by the Raw dataset. The packed inputs then undergo the enhancement network $F_{en-Raw}(\cdot)$, and finally unpacked by P^{-1} , which reshapes the twelve-channel output into a three-channel RGB image.

For the RGB format LLIE task, the enhancement process is simplified by removing the preprocessing and postprocessing operations of the Raw LLIE pipeline illustrated in Fig. 1(a), including P , β , γ , and P^{-1} . The proposed RGB format LLIE network, denoted as $F_{en-RGB}(\cdot)$, directly

takes in three-channel RGB images and outputs three-channel enhanced RGB images, which can be described as

$$I_{en-RGB} = F_{en-RGB} (I_{low-RGB}). \quad (2)$$

B. InDNet Architecture

In this section, we aim to improve the detail and color reconstruction performance of LLIE networks by deploying our proposed InD modules. Therefore, we take a classic U-net [47] as the baseline network architecture and equip it with InD modules. The U-net consists of an encoder part, a decoder part, and concatenate skip connections (CSCs) relating feature maps between these two parts. As illustrated in Fig. 1(b), we plan to insert one InD module before the max pooling operation of each stage in the encoder part. Besides, we introduce a global skip connection from the first convolution to the last convolution of the network by an addition operation and name it residual skip connection (RSC). We refer to the complete network as an InDNet. Note that when trained with different data formats, the network structure can be adjusted accordingly. Precisely, when given packed four-channel Bayer Raw data, we set the channel number of the first convolution to be 12 to match the output channel number as the RSC is required. On the other hand, due to the natural matching of the number of input and output channels of RGB inputs, the convolution before the first stage of the encoder is not required.

C. InD Module

To achieve efficient feature extraction that integrates both global and detailed information for better detail recovery and color reconstruction, in this section, we design an InD module to calculate unique global information values for each pixel of each channel. To avoid the vast computational cost brought by calculating intense pixel-to-pixel relations like

the NL neural network [17] does, the proposed InD module integrates the spatial-wise and channel-wise global information separately using SNL structures [18]. As illustrated in Fig. 1(c), the channel-wise global information in the left part is spatial-intensive and the spatial-wise global information in the right part is channel-intensive. Thus, we can diffuse the spatial-wise and channel-wise global information to each other by a matrix multiplication operation, resulting in a pixel-and-channel intensive global information matrix, which has the same shape as the input feature map. The InD module can be defined as

$$z_{ij} = \left(\delta \left(\sum_{i=1}^{N_p} \alpha_i \cdot x_i \right) \right) \otimes \left(\delta' \left(\sum_{j=1}^{N_c} \zeta_j \cdot x_j \right) \right) + x_{ij} \quad (3)$$

where

i) i and j denote the spatial and channel-wise index of each pixel of the input feature map, respectively.

ii) $\sum_{i=1}^{N_p} \alpha_i \cdot x_i$ denotes the spatial information integration operation which groups the features of all positions together via weighted averaging with weight α_i to obtain the spatial-wise global information, and α_i is achieved by an 1×1 convolution followed by a Softmax operation, as shown in the right part of Fig. 1(c).

iii) Correspondingly, $\sum_{j=1}^{N_c} \zeta_j \cdot x_j$ denotes the channel information integration operation which groups the features of all channels together via weighted averaging with weight ζ_j to obtain the channel-wise global information, and ζ_j is achieved by a global average pooling (GAP) operation followed by a Softmax operation, as shown in the left part of Fig. 1(c).

iv) $\delta(\cdot)$ and $\delta'(\cdot)$ denote the feature transform operation, which consists of two 1×1 convolutions with channel reduction ratio equals four.

v) $\otimes$ denotes the matrix multiplication operation, which merges the spatial-wise and channel-wise global information. We call this operation information diffusion.

D. Loss Function

Following [48], we use a joint loss function consisting of l_1 loss and MS-SSIM loss to train the proposed InDNet in a supervised manner. The joint loss function can be expressed as

$$\mathcal{L}_{\text{en}} = \lambda \mathcal{L}^{l_1}(I_{\text{gt}}, I_{\text{en}}) + (1 - \lambda) \mathcal{L}^{\text{ssim}}(I_{\text{gt}}, I_{\text{en}}) \quad (4)$$

where I_{gt} and I_{en} represent reference images and output enhanced images, respectively. $\lambda \in [0, 1]$ is the weight to balance the two losses.

IV. RGB-TO-RAW CUDA FRAMEWORK

This section provides details of our proposed RGB-to-Raw CUDA framework. First, we introduce the overall pipeline to offer a brief understanding of the workflow. Then, we present the design of every component, including the IFCN, the DATN, and the pretrained Raw-InDNet for unsupervised RGB format LLIE. The loss functions involved will be described accordingly.

A. Pipeline for CUDA Framework

Our CUDA framework is designed to bridge the Raw-and-RGB domain gap and resolve the unsupervised RGB LLIE task with the Raw domain prior inherent in the pretrained network. Note that, in this article, we define the Raw format Sony subset of SID [10] dataset as the source domain data and low-light images with RGB format as the target domain data. As illustrated in Fig. 2, the framework consists of three stages. In Stage I, the input low-light RGB images of the target domain, denoted as $I_{\text{RGB-low}}^T$, are taken by IFCN to achieve RGB-to-Raw format conversion, the converted output is denoted as $\hat{I}_{\text{Raw-low}}$. In Stage II, we propose DATN to reduce feature distribution distance between the converted Raw data $\hat{I}_{\text{Raw-low}}$ from the RGB domain and the Raw data I_{Raw}^S from source domain, resulting domain transferred $\hat{I}_{\text{Raw-trans}}$. DATN is composed of an iterative distribution transfer network (DTN) and a domain adversarial classifier (DAC). In the inference phase, the output of DTN, i.e., $\hat{I}_{\text{Raw-trans}}$, can be regarded as the output of DATN and input to the subsequent enhanced network. In the training phase, $\hat{I}_{\text{Raw-trans}}$ will be sent to DAC for domain adversarial classification to improve the distribution transfer capability of DTN. In Stage III, the domain transferred $\hat{I}_{\text{Raw-trans}}$ are enhanced by the pretrained InDNet, denoted as $F_{\text{en-Raw}}(\cdot)$, to obtain the final output $\hat{I}_{\text{RGB-normal}}$. Thus, the pipeline can be formulated as follows:

$$\hat{I}_{\text{RGB-normal}} = F_{\text{en-Raw}} \left(\text{DATN} \left(\text{IFCN} \left(\underbrace{I_{\text{RGB-low}}^T}_{\hat{I}_{\text{Raw-low}}} \right) \right) \right) \quad (5)$$

B. RGB-to-Raw Format Conversion Network

Due to the lack of a standard RGB-to-Raw conversion process, we use a simple U-Net [47] as our IFCN to tackle the image format disparity, as illustrated in Fig. 2(a). IFCN takes RGB format data I_{RGB} as input and outputs converted Raw data $\hat{I}_{\text{Raw}}$, which can be defined as

$$\hat{I}_{\text{Raw}} = \text{IFCN}(I_{\text{RGB}}). \quad (6)$$

During the training phase, in order to match the input form of the subsequent Raw LLIE network, we follow [10] to preprocess the reference four-channel Raw images $I_{\text{Raw-4c}}$ from the original one-channel Raw images $I_{\text{Raw-1c}}$, which can be expressed as follows:

$$I_{\text{Raw-4c}} = \mathcal{P}(I_{\text{Raw-1c}}) - \beta \quad (7)$$

where $\mathcal{P}$ represents the packing operation and β is the black level as described in (1). We use l_1 loss to train IFCN, which can be described as

$$\mathcal{L}_{\text{IFCN}} = \mathcal{L}^{l_1}(\hat{I}_{\text{Raw}}, I_{\text{Raw-4c}}) \quad (8)$$

where $\hat{I}_{\text{Raw}}$ represents the converted Raw format data and $I_{\text{Raw-4c}}$ represents the reference packed Raw data.

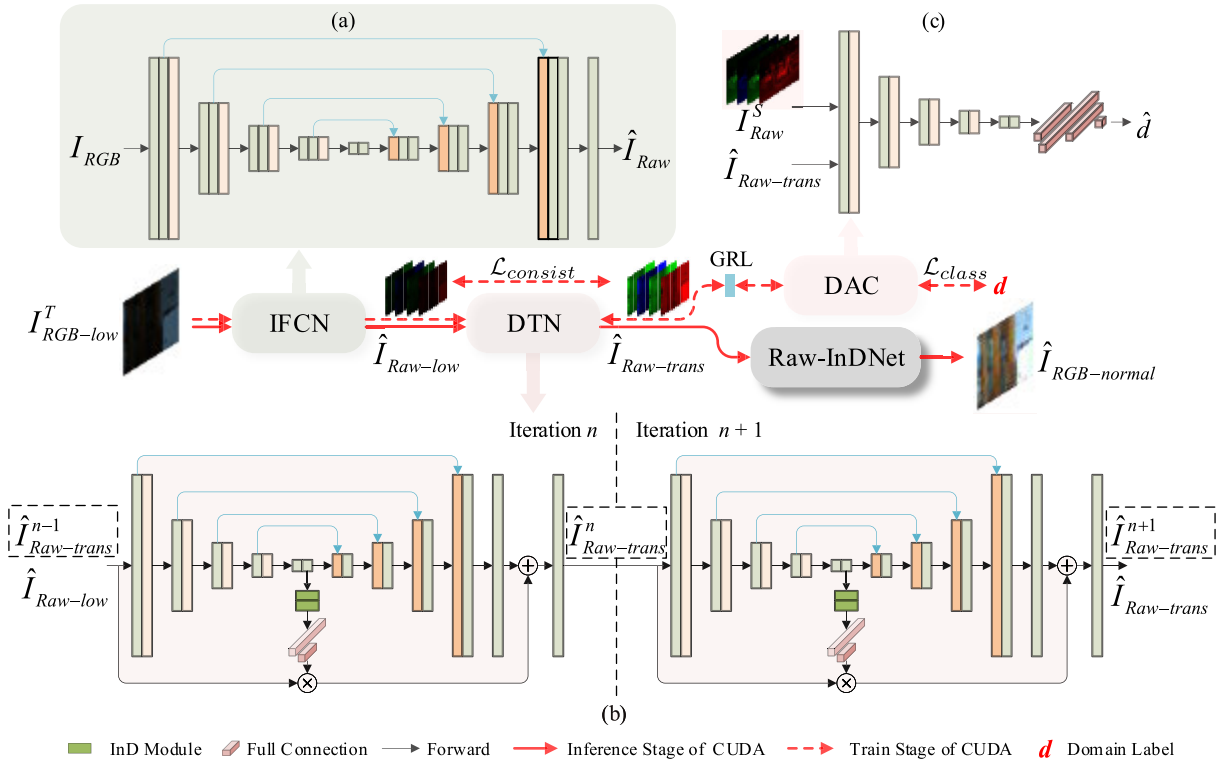


Fig. 2. Detailed illustration of our proposed CUDA framework. In order of the process, the framework includes an IFCN (a) to convert RGB domain low-light input $I_{RGB-low}^T$ to Raw domain image $\hat{I}_{Raw-low}$, an iterative DTN (b) for $\hat{I}_{Raw-low}$ to obtain domain transferred $\hat{I}_{Raw-trans}$, a DAC (c) to achieve unsupervised training for DTN and a Raw-InDNet to make the final LLIE for $\hat{I}_{Raw-trans}$, resulting $\hat{I}_{RGB-normal}$ as output of the framework. Note that in (b), $\hat{I}_{Raw-trans}^0 = \hat{I}_{Raw-low}$ when $n = 0$. GRL represents the gradient reversal layer, $\otimes$ denotes matrix multiplication, and $\oplus$ denotes element-wise addition.

Note that we train the IFCN with normal-light Raw and RGB image pairs in the source domain instead of low-light ones to ensure adequate effective information retention. In addition, when used as a format converter for the CUDA framework, its parameters are fixed.

C. Domain Adversarial Transfer Network

With different data formats and different data distributions, LLIE models trained on the Raw dataset are not applicable to RGB images of the target domain. Therefore, the prior information contained in the Raw domain network cannot be exploited. In Section IV-B, we propose to resolve the format difference by IFCN. In this section, we design an unsupervised DATN to reduce the data distribution differences between the target and source domains through an adversarial training strategy. Given a converted target domain Raw image generated by IFCN, denoted as $\hat{I}_{Raw-low}$, DATN can slightly change its distribution to match the distribution of the source domain while keeping its format and effective visual information unchanged, the domain transferred image can be defined as

$$\hat{I}_{Raw-trans} = \text{DATN}(\hat{I}_{Raw-low}). \quad (9)$$

The architecture of the proposed DATN is illustrated in Fig. 2, it includes a deep iterative DTN as in Fig. 2(b) and a DAC as in Fig. 2(c).

1) Iterative DTN for DATN: As the distribution transfer part of DATN for the LLIE task, we aim to manipulate the data distribution of the target domain to match the distribution of the source domain while keeping as much of the original image features as possible. Therefore, in order to achieve better low-level visual feature variation, we use a modified U-net structure as the baseline to design our DTN. Furthermore, we notice that in Raw domain training, the input Raw data is magnified by exposure time ratio γ before being sent into Raw-InDNet as described in (1). This leads to a huge overall numerical difference in the distribution compared with the target domain data. To address this overall numerical difference along with other possible distribution differences, we design the DTN to perform an affine transformation (AT) on the input data. The AT includes an image-level multiplicative operation and a pixel-level additive operation. Specifically, as illustrated in Fig. 2(b), given a converted target domain Raw image $\hat{I}_{Raw-low}$ as input, we apply two InD modules after the bottleneck of the U-shaped DTN to get the global-wise multiplicative coefficients γ' . Meanwhile, the pixel-wise additive coefficients β' are obtained from the last convolution of DTN. To make it clear, the transferred Raw output $\hat{I}_{Raw-trans}$ can be described as

$$\hat{I}_{Raw-trans} = \gamma' \times \hat{I}_{Raw-low} + \beta'. \quad (10)$$

In addition, the AT defined in (10) can be applied iteratively to enable higher order DTN for more sophisticated gain control,

as illustrated in Fig. 2(b). Specifically, the n th order DTN can be defined as

$$\hat{I}_{\text{Raw-trans}}^n = \gamma' \times \hat{I}_{\text{Raw-trans}}^{n-1} + \beta' \quad (11)$$

where n is the number of iterations and $\hat{I}_{\text{Raw-trans}}^0 = \hat{I}_{\text{Raw-low}}$.

Note that the affine coefficients γ' and β' are shared and reused during the iteration process. Therefore, we can adjust the number of iterations through a simple gain control strategy during the testing process to avoid underexposed or overexposed outcomes, even if n is fixed during the training process. One of the most straightforward gain control strategies is to set a mean threshold T and stop iteration when the output mean of the current iteration surpasses the threshold T . We refer to this strategy as mean gain control (MGC).

2) DAC for DATN: In this section, we use an adversarial training strategy to achieve unsupervised distribution transfer. As illustrated in Fig. 2, a domain classifier is connected to DTN via a gradient reversal layer (GRL) [42]. The GRL multiplies the gradient by a negative constant during the backpropagation-based training. We refer to this domain classifier as DAC, the detailed structure of which is illustrated in Fig. 2(c). To effectively distinguish the global-wise distribution difference between the source and the target domain images, we apply four residual convolution blocks to reduce the feature size and integrate global-wise discriminative information. Then, we use several fully connected layers to map this information to 2-D category vectors, which represent the source and target domain, respectively. The GRL ensures that the feature distribution over the two domains is made similar for DTN while as indistinguishable as possible for DAC.

3) Loss Functions for Training DATN: Dissimilar to high-level visual processing tasks, the low-level vision tasks, here refer to the LLIE task, regard pixel restoration as the primary objective. Consequently, the proposed DTN is designed to avoid altering the fundamental attributes of target domain images, including color and texture information. Therefore, we set the training proceeds to minimize the feature consistency loss between the normalized input and output of DTN for target domain examples only. The feature consistency loss is defined as

$$\mathcal{L}_{\text{consist}} = \mathcal{L}^{l_1+\text{ssim}} \left(\text{LN} \left(\hat{I}_{\text{Raw-trans}} \right), \text{LN} \left(\hat{I}_{\text{Raw-low}} \right) \right) \quad (12)$$

where LN means layer normalization and $\mathcal{L}^{l_1+\text{ssim}}$ represents a joint loss function consisting of l_1 loss and MS-SSIM loss, which is the same as described in (4).

For domain binary classification training of DAC, we use cross-entropy loss

$$\mathcal{L}_{\text{class}} = \mathcal{L}^{\text{cross-entropy}} \left(\text{DAC} \left(\hat{I}_{\text{Raw}} \right), d \right) \quad (13)$$

where $\mathcal{L}^{\text{cross-entropy}}$ represents cross-entropy loss, $\hat{I}_{\text{Raw}}$ represents the input Raw images of DAC, it can be source domain Raw images I_{Raw}^s from SID dataset or DTN-transferred Raw images $\hat{I}_{\text{Raw-trans}}$ from target domain. d denotes the domain label of input Raw images, 1 for the source domain and 0 for the target domain. The overall loss function for training DATN is thus written as

$$\mathcal{L}_{\text{DATN}} = \mathcal{L}_{\text{consist}} + \mathcal{L}_{\text{class}}. \quad (14)$$

4) Differences With Domain Adaptation for High-Level Visual Tasks: It is worth mentioning that many previous works on domain adaptation, such as DANN [49], mainly focus on learning discriminative and domain-invariant features for subsequent high-level vision tasks. Nevertheless, our practical experience suggests that the specific pixel-level recovery demands of low-level vision tasks restrict the acquisition of abstract high-level domain-invariant representations. In other words, in the scenario of this article, the abstract domain-invariant representations are unable to fulfill the input requirements of subsequent LLIE networks. Also, training the DATN and subsequent LLIE networks jointly is difficult. Based on the above arguments, in this section, we propose to keep the well-trained subsequent Raw-InDNet fixed and train DATN to slightly change the data distribution of the target domain to match that of the source domain. The benefit of this strategy lies in its capacity to perform fast domain adaptation for any unseen RGB domain datasets, as only the lightweight DATN needs training or fine-tuning.

D. Unsupervised RGB LLIE on Raw-InDNet

As described in Section IV-C4, in order to avoid the instability introduced by joint training subsequent LLIE network for hard-to-obtain domain-invariant representations, we propose to keep the parameters of a well-trained Raw domain LLIE network fixed and utilize it as the predictor of DATN. In this article, we take the Raw domain trained InDNet described in Section III as the predictor of DATN and call it Raw-InDNet, the architecture of which can be seen in Fig. 1(b). Note that the Raw-InDNet is trained only on the SID Raw dataset of the source domain. Given IFCN-converted and DTN-transferred Raw format $\hat{I}_{\text{Raw-trans}}$ from the target domain, the final output of our proposed CUDA framework is

$$\hat{I}_{\text{RGB-normal}} = \mathcal{P}^{-1} \left(F_{\text{en-Raw}} \left(\hat{I}_{\text{Raw-trans}} \right) \right) \quad (15)$$

where $F_{\text{en-Raw}}(\cdot)$ denotes Raw-InDNet and $\mathcal{P}^{-1}$ means the unpacking operation, which reshapes the twelve-channel output into three-channel RGB images, as described in (1).

V. EXPERIMENTAL RESULTS

A. Datasets and Evaluations Metrics

In this section, we report the performance of the proposed InDNet and CUDA framework through qualitative and quantitative comparison. Since the comparison involves two image formats, Raw and RGB, and two learning strategies, supervised and unsupervised learning, we outline the specific verification procedure including the datasets and evaluation criteria as follows.

For single format supervised trained InDNet, we plan to assess its performance on both formats of datasets, the most representative of which are the SID [10] dataset with Raw format and LOL [5] dataset with RGB format. The SID dataset consists of two subsets collected using two different sensors, i.e., Sony $\alpha 7S$ II with a Bayer color filter array and Fuji X-T2 with an X-Trans color filter array. Totally 5094 short-exposure Raw input images and 424 corresponding long-exposure reference images were collected. We plan to

follow [10] and evaluate the performance on both Sony and Fuji subsets in terms of point SNR (PSNR) and structural similarity index measure (SSIM) [50]. PSNR and SSIM are the two most widely used reference image quality assessment (IQA) methods in the LLIE field. PSNR evaluates the fidelity between the enhanced image and the reference by calculating the mean square error (MSE) between two images. SSIM is a perception-based metric that measures the structural similarity between two images. Unlike PSNR, which only considers pixel-wise differences, SSIM takes into account structural information, luminance, and contrast, providing a more comprehensive assessment of image quality. The LOL dataset is a real captured dataset composed of 500 pairs of low and normal-light RGB images obtained from real scenes, in which 485 pairs are used for training and the remaining 15 pairs are used for testing. Following [9], three metrics are adopted for comprehensive quantitative comparison on the LOL dataset. In addition to PSNR and SSIM, we also include learned perceptual image patch similarity (LPIPS) [51]. LPIPS utilizes deep learning to directly learn feature representations from data and measures perceptual similarity between images by comparing their feature representations across multiple scales. This offers improved alignment with human perception in terms of IQA compared to PSNR and SSIM. By integrating PSNR, SSIM, and LPIPS, our proposed LLIE model can undergo a thorough and accurate assessment of its performance.

For the evaluation of the CUDA framework for unsupervised RGB format LLIE, we plan to compare it with other unsupervised methods on the LOL dataset using metrics including PSNR, SSIM, and LPIPS. Besides, to illustrate the generalization performance of the CUDA framework, we further perform an evaluation on the real part of LOL-v2 [52] dataset, which is larger and more diverse than LOL, including 689 low and normal-light pairs for training and 100 pairs for testing.

B. Implementation Details

As illustrated in Fig. 1(b), a total of 5 InD modules are inserted into the baseline U-net [10], [47] architecture to form our InDNet. For a fair comparison, we train InDNet for Raw images with the same image processing pipeline as in [10]. We train InDNet with (4) for 4000 epochs, and λ in (4) is set to 0.84. During training, Adam optimizer is used and the learning rate is initially set to $1e^{-4}$, then decreased to $1e^{-5}$ after 2000 epochs. We crop random 512×512 patches from Raw images and flip, rotate, or transpose it randomly as the input for training the network. The full images are taken as input in the testing phase. For the training on the LOL dataset, we discard the first convolution layer and change the output channel of InDNet to three. The training patch is set to 256×256 . Other settings for training RGB images remain the same as for training Raw images.

As for the implementation of the CUDA framework, we train the three stages of the framework separately. First of all, the Raw-InDNet of the framework is exactly the InDNet trained on the SID Sony subset, the parameters of which are fixed when used as the predictor of the framework. IFCN is also pretrained with the same hyperparameters used for

TABLE I
QUANTITATIVE COMPARISON BETWEEN THE PROPOSED RAW-INDNET AND OTHER STATE-OF-THE-ART SUPERVISED METHODS IN TERMS OF PSNR/SSIM ON THE SID DATASET. THE BEST RESULTS ARE HIGHLIGHTED IN **BOLD**, AND THE SECOND-BEST RESULTS ARE IN UNDERLINE. “-” INDICATES THAT EVALUATIONS ARE NOT AVAILABLE

Methods	Sony	Fuji
	PSNR/SSIM	PSNR/SSIM
MCN [15]	29.22/0.778	26.89/0.699
LDC [13]	29.56/ 0.799	27.18/0.703
EEMEFN [12]	29.60/0.795	27.38/ <u>0.723</u>
GIA-Net [14]	29.72/0.795	<u>28.15</u> /0.722
ALEN [16]	<u>29.86</u> /0.787	-
REENet [11]	29.17/0.792	27.29/0.720
MCR [54]	29.65/ <u>0.797</u>	-
SID (Baseline) [10]	28.88/0.787	26.61/0.680
Raw-InDNet (Ours)	30.38 /0.796	28.54 / <u>0.738</u>

training Raw-InDNet. The learned parameters are fixed when used as the format converter of the CUDA framework. We train DATN with (14) for 400 epochs on the LOL dataset, only using the low-light images. The iteration number n in (11) is set to three during training and remains unchanged when reporting the unsupervised results without MGC. When reporting the results with MGC, the maximum number of iterations N is set to four, and the mean threshold T is set to 0.18. Due to the existence of GRL, the adversarial training of DTN and DAC can be completed simultaneously, and then we conduct an additional DTN feature consistency training using (12) in each epoch. The learning rate of Adam optimizer for DTN is initially set to $1e^{-4}$, then decreased to $1e^{-5}$ after 200 epochs, and the learning rate for DAC is set to $1e^{-5}$. Other preprocessing strategies and training hyperparameters are consistent with the supervised training. For the evaluation of the LOL-v2 dataset, we finetuned the DATN trained on the LOL dataset with the learning rate set to $1e^{-5}$ for 50 epochs without reference. We also report both the non-MGC results with fixed $n = 3$ and the with-MGC results with $N = 8$, $T = 0.1$.

All experiments in this article are conducted on an i7 3.6 GHz CPU and an Nvidia GTX 1080Ti GPU using TensorFlow [53] framework.

C. Experimental Results for Supervised InDNet

1) *Supervised Results on Raw Datasets*: In this section, we compare our proposed method with several most recent state-of-the-art methods on the SID dataset, including SID [10] as the baseline, EEMEFN [12], LDC [13], ALEN [16], GIA-Net [14], MCN [15], REENet [11], and MCR [54]. For all the methods mentioned above, we obtain the quantitative results by taking the reported results from the original papers or by running publicly released codes. As for qualitative comparison, we reproduce GIA-Net [14] and use the pretrained model of SID [10], and ALEN [16] to achieve their visual results.

The quantitative comparison results are illustrated in Table I. It can be seen that our InDNet outperforms the second-best results by 0.52 and 0.39 dB on the Sony and Fuji

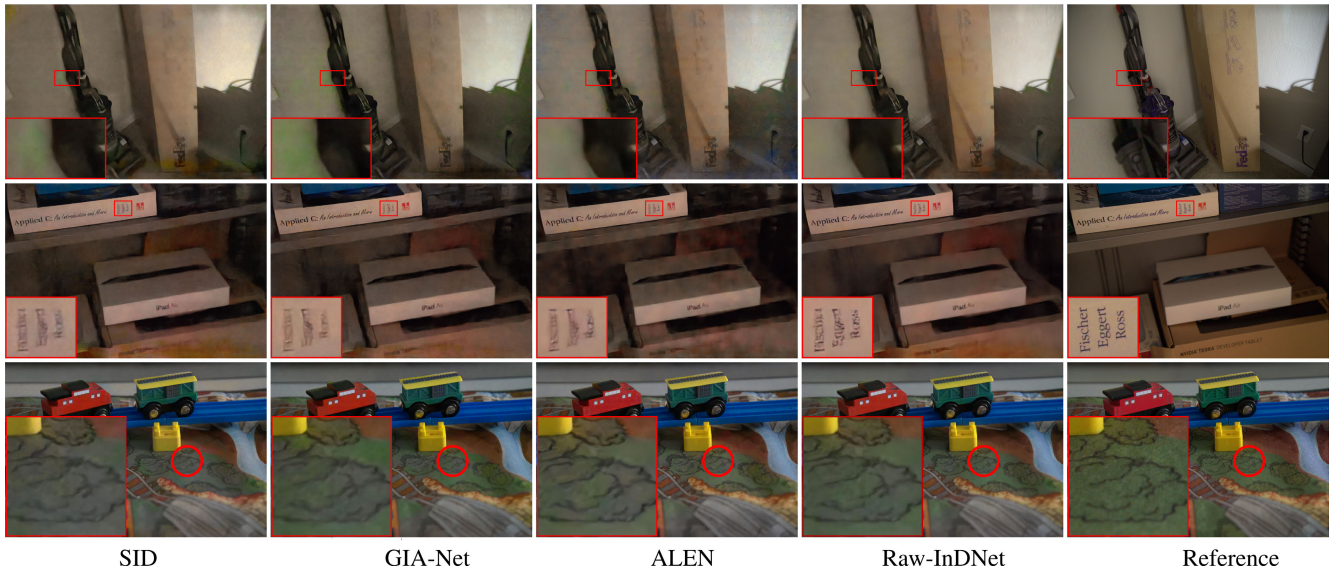


Fig. 3. Qualitative comparison with state-of-the-art supervised methods on the SID Sony subset. Please zoomed-in view for the best view. Note that we selected methods with the top three PSNR results in Table I for visual comparison.

subsets, respectively, in PSNR, and achieves the best SSIM result as 0.738 on the Fuji subset, which outperforms the second-best SSIM result by 0.015. Fig. 3 shows the qualitative comparison results. Note that Fig. 3 is a reused version of the visual comparisons in our previous published paper [19]. It can be observed from the first row that our InDNet achieves the best color consistency and accuracy. In the second row, we compare the detail-recovery capabilities. Although none of us succeeded in fully recovering the letters, clearer lines and edges can be seen from the enhanced image of InDNet. In the third row, we can see that the green tree-like carpet highlighted in the image generated by InDNet has the most similar colors and textures to those in the reference. In general, the above comparison fully illustrates the effectiveness of our InDNet in detail reconstruction and color restoration.

2) Supervised Results on RGB Dataset: In order to verify the performance of our proposed InDNet on RGB format LLIE task, we conduct experiments with InDNet on the LOL dataset and compare the results with several most recent state-of-the-art LLIE methods for RGB images, including LIME [24], RetinexNet [5], KinD [6], KinD++ [7], MIRNet [25], HWMNet [26], IAT [8], LLFlow [9], and Bread [55].

The quantitative comparison results on the LOL dataset are illustrated in Table II. The results show that our InDNet achieves the best PSNR result at 24.44 dB, outperforming the second-best result by 0.2 dB. We also obtained the most competitive SSIM and LPIPS results as 0.855 and 0.119, respectively, tied for first place with LLFlow. Fig. 4 shows the qualitative comparison results. It can be observed from the first row that our InDNet achieves the best detail-recovery and noise suppression result. In the second row, we compare the color accuracy of the enhanced images. Please pay attention to the difference between the red, blue, and black in the red box in the lower right corner and the colors in the reference image, obviously, our InDNet obtains the most accurate color representation. In the third row, we can see that our InDNet

TABLE II

QUANTITATIVE COMPARISON BETWEEN OUR PROPOSED RGB-INDNET AND OTHER STATE-OF-THE-ART SUPERVISED METHODS IN TERMS OF PSNR/SSIM/LPIPS ON THE LOL DATASET. THE BEST RESULTS ARE HIGHLIGHTED IN **BOLD**, AND THE SECOND BEST RESULTS ARE IN UNDERLINE. $\uparrow(\downarrow)$ DENOTES THAT LARGER (SMALLER) VALUES ARE BETTER. NOTE THAT WHEN DISPLAY VALUES ARE THE SAME, THE RANKING IS DETERMINED BY MORE PRECISE DECIMAL DIGITS IF AVAILABLE

Methods	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$
LIME [24]	16.76	0.564	0.350
RetinexNet [5]	16.77	0.462	0.474
KinD [6]	20.87	0.802	0.207
KinD++ [7]	21.30	0.823	0.175
MIRNet [25]	24.10	0.847	0.131
HWMNet [26]	<u>24.24</u>	0.852	0.120
IAT [8]	23.38	0.809	0.216
LLFlow [9]	21.15	<u>0.855</u>	0.119
Bread [55]	22.96	0.838	0.160
SID (Baseline) [10]	22.48	0.835	0.141
RGB-InDNet (Ours)	24.44	0.855	<u>0.119</u>

achieves the closest visual perception to the reference image, both in terms of color and detail accuracy. The color and texture details of SID, IAT, and MIRNet are not restored well, while HWMNet shows obvious over-brightness compared with the reference. In general, the above comparison fully illustrates the effectiveness of our InDNet in detail reconstruction and color restoration.

D. Unsupervised Results for RGB-to-Raw CUDA Framework on RGB Datasets

In this section, we report the performance of our CUDA framework and compare it with several state-of-the-art unsupervised LLIE methods on the LOL and LOL-v2 dataset, including Zero-DCE [29], EnlightenGAN [30], and RUAS [31]. For intuitive comparison, we also include the semi-supervised method Sharma and Tan [56] and two other

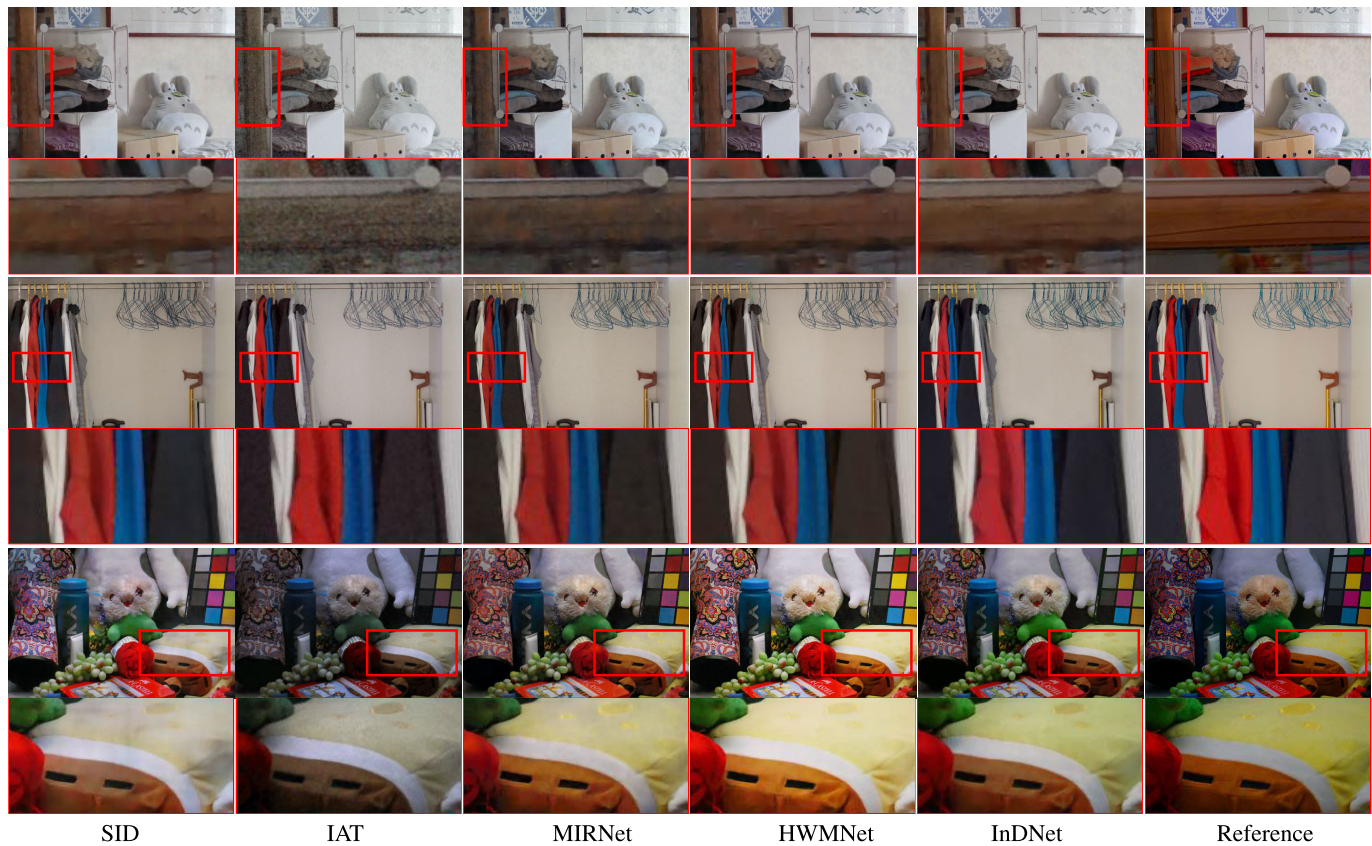


Fig. 4. Qualitative comparison with state-of-the-art supervised methods on the LOL dataset. Please zoomed-in view for the best view. Note that we selected methods with the top four reported PSNR results in Table II for visual comparison here.

TABLE III

QUANTITATIVE COMPARISON BETWEEN THE CUDA FRAMEWORK AND OTHER STATE-OF-THE-ART UNSUPERVISED METHODS USING PSNR, SSIM, AND LPIPS ON LOL AND LOL-v2 DATASETS. WE ALSO INCLUDE SEVERAL SUPERVISED AND SEMI-SUPERVISED METHODS FOR INTUITIVE COMPARISON. THE BEST RESULTS ARE HIGHLIGHTED IN **BOLD**, AND THE SECOND BEST RESULTS ARE IN UNDERLINE. “—” INDICATES THE RESULTS ARE NOT AVAILABLE

	Methods	LOL			LOL-v2		
		PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$
<i>Supervised</i>	LIME [24]	16.76	0.564	0.350	15.24	0.470	—
	RetinexNet [5]	16.77	0.428	0.474	16.10	0.409	0.543
<i>Semi-supervised</i>	Sharma <i>et al.</i> [56]	16.88	0.670	0.315	18.34	0.640	—
	Zero-DCE [29]	14.86	0.562	0.335	18.06	0.577	0.313
<i>Unsupervised</i>	EnlightenGAN [30]	17.46	0.655	<u>0.323</u>	18.63	0.679	<u>0.310</u>
	RUAS [31]	18.23	0.717	0.354	18.84	0.512	0.299
	CUDA (ours)	<u>20.11</u>	<u>0.755</u>	0.381	<u>19.14</u>	<u>0.718</u>	0.488
	CUDA-MGC (ours)	20.42	0.761	0.376	20.49	0.752	0.470

classic supervised methods, LIME [24] and RetinexNet [5]. Note that since we are the first work to implement UDA for cross-format LLIE tasks, we do not compare the results with other domain adaptation methods.

As shown in Table III, our CUDA framework, with or without the adjustment of GMC, achieves the best PSNR and SSIM results on both LOL and LOL-v2 datasets. Also, with the application of MGC, we obtain obvious performance gains, especially on the LOL-v2 dataset. The possible reason is that the complex scenes and illumination conditions of LOL-v2 require a more adaptive gain strategy. As for LPIPS value, the CUDA framework obtains not good enough but still competitive results, the possible reason is that the Raw-InDNet used as the predictor of the proposed unsupervised framework

is trained on the extreme dark images, thus the enhanced results tend to be over-denoised, resulting relatively poor LPIPS results. More optimization of the predictor needs to be done in future work. It is worth noting that even without using the empirical hyperparameters of MGC, our unsupervised framework still outperforms state-of-the-art supervised and semi-supervised methods including LIME [24], RetinexNet [5] and Sharma and Tan [56].

Fig. 5 shows the qualitative comparison results on LOL and LOL-v2 datasets. We also include the supervised method RetinexNet [5] for intuitive comparison. The first set of results demonstrates the overall low-light enhancement and noise suppression abilities of the proposed CUDA framework, as we can see that CUDA and CUDA-MGC achieve the best texture

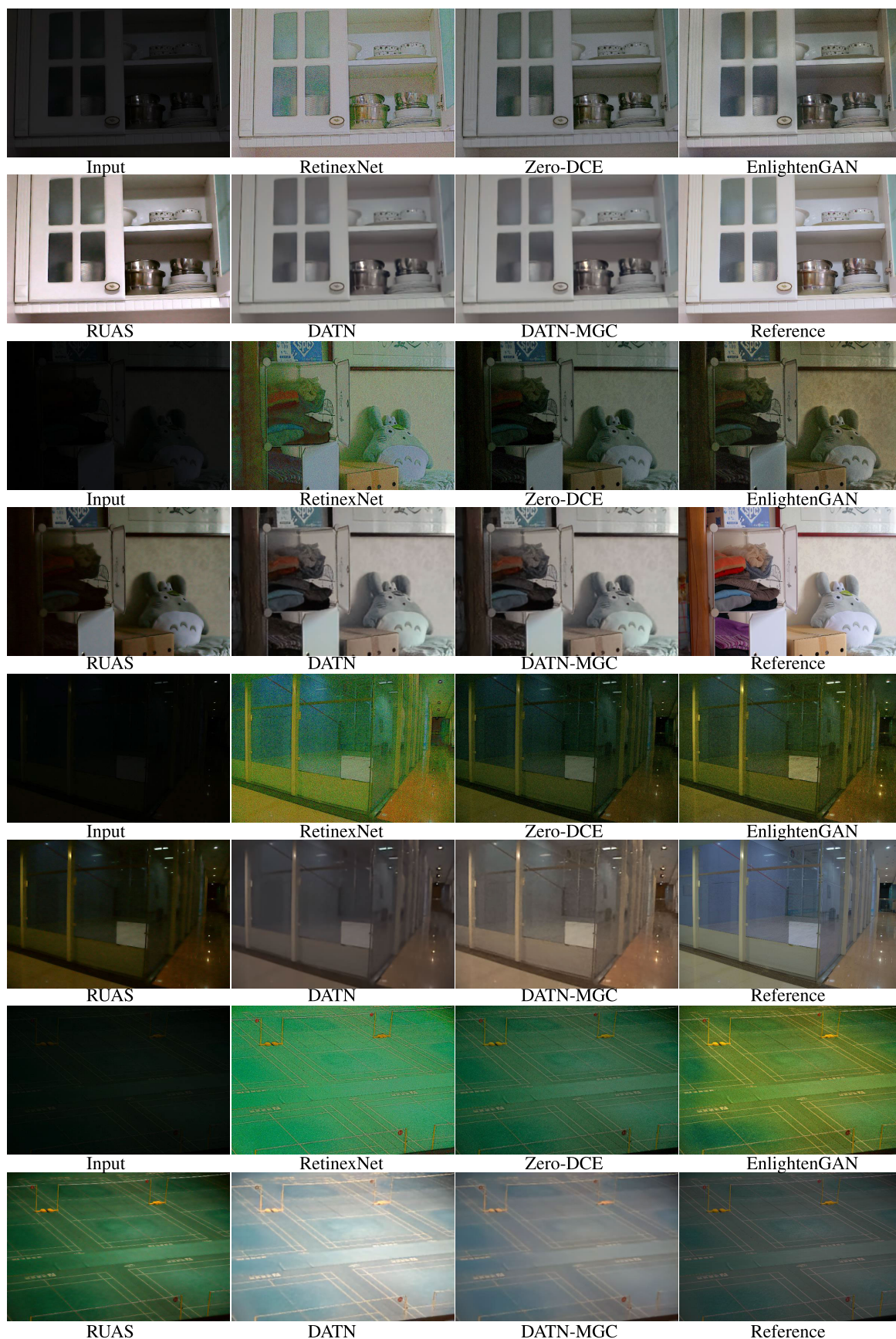


Fig. 5. Qualitative comparison with state-of-the-art unsupervised methods on LOL and LOL-v2 dataset. Please zoomed-in view for the best view. For intuitive comparison, we also include the classic supervised method RetinexNet. The first two sets of images are from the LOL dataset and the last two sets of images are from the LOL-v2 dataset.

TABLE IV

ABLATION STUDY OF INDNET COMPONENTS IN TERMS OF PSNR/SSIM. THE BEST RESULTS ARE HIGHLIGHTED IN **BOLD**, AND THE SECOND BEST RESULTS ARE IN UNDERLINE

baseline	SNL	InD	RSC	$\mathcal{L}^{l_1}$	$\mathcal{L}^{ssim}$	Sony PSNR/SSIM	LOL PSNR/SSIM	Params
✓				✓		28.88/0.787	22.48/0.835	7.76M
✓			✓	✓		28.96/0.784	22.54/0.840	7.77M
✓				✓	✓	29.06/ <u>0.790</u>	22.92/ <u>0.845</u>	7.76M
✓	✓			✓		29.76/0.788	23.04/0.829	7.98M
✓		✓		✓		<u>30.13/0.790</u>	<u>23.07/0.838</u>	8.11M
✓		✓	✓	✓	✓	30.38/0.796	24.44/0.855	8.12M

and color restoration results. Results in Sets No.2 and No.3 demonstrate the enhanced capabilities of CUDA for extreme low-light scenes, where other methods either suffer from severe noise or fail to recover details. In the last set, CUDA and CUDA-MGC clearly achieve the best global color restoration results. It is worth noting that the last two sets of results show the difference in overall brightness between CUDA and CUDA-MGC. We can see that the result of CUDA-MGC in set No.3 is brighter while it is darker in the last set, but they both are closer to the references than the results of CUDA, which demonstrates the effectiveness of MGC. We can also observe that the MGC does not affect the results of the first two comparison groups, which shows the rationality of MGC, as it only makes adjustments when necessary.

VI. ABLATION STUDY AND ANALYSIS

A. Ablation Study for InDNet

To thoroughly validate the effectiveness of our proposed InDNet architecture, we conduct several ablation experiments to demonstrate its performance impact on the Sony subset of SID and the LOL dataset. For simplicity, we only evaluate the performance using PSNR and SSIM.

1) *Effectiveness of InDNet Components*: We first train SID [10] with l_1 loss as our baseline. On this basis, we add other components, such as the RSC and the MS-SSIM loss for network training. In this study, we insert five InD modules into the baseline network, as it reports the best results. We also compare our InD module with the SNL structures [18] where we get inspiration from, five SNL modules are inserted into the baseline network at the same location as the InD modules.

Table IV lists the evaluation results and the number of parameters of each network. It can be seen that with RSC operation and MS-SSIM loss, minor performance gains are obtained on the Sony subset, while on the LOL dataset, their gain effects are significantly higher. The possible reason is that the image resolution of the LOL dataset is much lower than that of the Sony subset, thus the RSCs and MS-SSIM loss play more important roles in detail recovery. The proposed InD modules significantly improve the PSNR results of the baseline network, i.e., 1.25 dB (from 28.88 to 30.13 dB) on the Sony subset and 0.59 dB (from 22.48 to 23.07 dB) on the LOL dataset. It is worth noting that the network inserted with InD modules also outperforms the network inserted with SNL modules in terms of PSNR value by 0.37 dB on the Sony subset and by 0.09 in terms of SSIM on the LOL dataset, respectively,

TABLE V

ABLATION STUDY OF IND MODULE IMPLANTATION IN TERMS OF PSNR/SSIM. THE BEST RESULTS ARE HIGHLIGHTED IN **BOLD**, AND THE SECOND BEST RESULTS ARE IN UNDERLINE

schedule	PSNR	SSIM	Params
[0, 0, 0, 0, 0, 0, 0, 0, 0]	29.06	0.790	7.760M
[1, 0, 0, 0, 0, 0, 0, 0, 0]	29.58	0.791	7.761M
[0, 1, 0, 0, 0, 0, 0, 0, 0]	29.82	0.793	7.765M
[0, 0, 1, 0, 0, 0, 0, 0, 0]	29.86	0.792	7.777M
[0, 0, 0, 1, 0, 0, 0, 0, 0]	29.99	0.793	7.827M
[0, 0, 0, 0, 1, 0, 0, 0, 0]	30.01	<u>0.794</u>	8.024M
[0, 0, 0, 0, 0, 1, 0, 0, 0]	29.83	0.792	7.827M
[0, 0, 0, 0, 0, 0, 1, 0, 0]	29.95	0.793	7.777M
[0, 0, 0, 0, 0, 0, 0, 1, 0]	29.88	0.792	7.765M
[0, 0, 0, 0, 0, 0, 0, 0, 1]	29.58	0.792	7.761M
[1, 1, 1, 1, 1, 0, 0, 0, 0]	30.24	0.795	8.113M
[1, 1, 1, 1, 1, 1, 1, 1, 1]	<u>30.22</u>	<u>0.794</u>	8.201M

showing better overall performance on high-resolution images and detail recovery capabilities on low-resolution images. This result proves the ability of our proposed InD module to extract global but refined information at the same time. At last, we can observe from the comparison of the last row that combined with RSC and MS-SSIM loss, the proposed InDNet achieves the state-of-the-art LLIE performance on both Raw and RGB datasets with less than 0.4M added parameters, showing the efficiency and effectiveness of the InD module.

2) *Number and Location Analysis of InD Module*: As a plug-and-play feature extraction module, the InD module can be inserted into any part of the convolutional network without changing the size or the shape of the features. Thus, we design this ablation study to discover the most effective amounts and locations to place InD Modules. As illustrated in Fig. 1(b), our baseline is a U-shaped network with four down-sampling stages, a bottleneck stage, and four up-sampling stages. We plan to insert one InD module into each stage, a total of nine InD modules for nine stages in sequence. We use a binary list to indicate whether the InD module is inserted at the nine corresponding positions. For example, schedule [0,0,0,0,1,0,0,0,0] represents only one InD module placed at the bottleneck stage. Note that the following experiments are only conducted on the Sony subset using PSNR and SSIM, the training loss is the joint loss of l_1 and MS-SSIM loss and RSCs are not applied.

It is obvious from Table V that different degrees of performance improvement can be obtained by inserting the proposed InD modules at different stages of the baseline network. Among all the locations, the bottleneck implantation outperforms others in terms of both PSNR and SSIM values. Correspondingly, with the most convolutional channels in the bottleneck, the most additional parameters are introduced. Moreover, the overall performance improvement shows an upside-down U-shape symmetrical with the bottleneck, in other words, implants closer to the network bottleneck obtain more performance gains, whereas implants closer to the input or output layers gain less improvement. It is also worth noting that the implantation at stages with the same feature size achieved similar performance gains, the possible reason is that CSCs enable the two stages to share cross-stage information.

TABLE VI

ABLATION STUDY OF CUDA FRAMEWORK COMPONENTS ON LOL DATASET IN TERMS OF PSNR/SSIM. THE BEST RESULTS ARE HIGHLIGHTED IN **BOLD**, AND THE SECOND BEST RESULTS ARE IN UNDERLINE

Schedule	PSNR	SSIM
w/ full DATN	20.11	0.755
w/o AT in DTN	17.07	0.633
w/o InD in DTN	<u>18.05</u>	<u>0.725</u>
w/o $\mathcal{L}_{consist}$	6.54	0.053
w/o DATN	8.48	0.277

Based on the above analysis and for efficiency consideration, we only implant InD modules at down-sampling stages and bottleneck stages to form our InDNet, and the up-sampling part maintains a full convolution state for better information reconstruction. The results in the last row of Table V also confirm that redundant InD implantation will not bring more performance gains, but will bring more parameters and will also lead to unstable convergence in practice.

B. Ablation Study for CUDA Framework

To thoroughly validate the effectiveness of our proposed CUDA framework, we conduct several ablation experiments to demonstrate its performance impact on the LOL dataset in terms of PSNR and SSIM values.

1) Effectiveness of CUDA Framework Components:

Table VI lists the experimental results of the ablation of different parts of CUDA, including the feature consistency loss $\mathcal{L}_{consist}$ of (12), the AT of (10) in DTN, the InD module in DTN, and the whole DATN branch. In addition, Fig. 6 shows the impact of ablating these parts on the training process. Note that w/o AT in the second row means that the DTN uses a simple U-shaped network rather than a U-shaped network combined with an AT.

As we can see, the last row reports the result without the DATN branch in the framework, it proves that even with the format conversion network IFCN and the subsequent enhancement network, the domain differences between different datasets still greatly affect the cross-domain enhancement performance. Therefore, the existence of DATN is necessary. The reason why the penultimate row of Table VI performs the worst results is that the missing of feature consistency constraints cause the output of DTN to remain zero and unable to perform gradient update, the black loss curve in Fig. 6 proves this point. The second and the third row of Table VI show different degrees of performance degradation compared to the results of the first row, which proves the effectiveness of the AT structure and the InD modules in DTN. Combined with the training convergence curve in Fig. 6, we believe that the AT structure in DTN has great impacts on the speed of training convergence and the domain adaptation performance. Also, InD plays an important role in integrating global information for DTN.

2) Analysis of the Role of DATN: In order to further prove the effectiveness of our proposed CUDA framework and analyze the role of DATN in it, we designed a series of ablation experiments and report the results in Table VII. We first

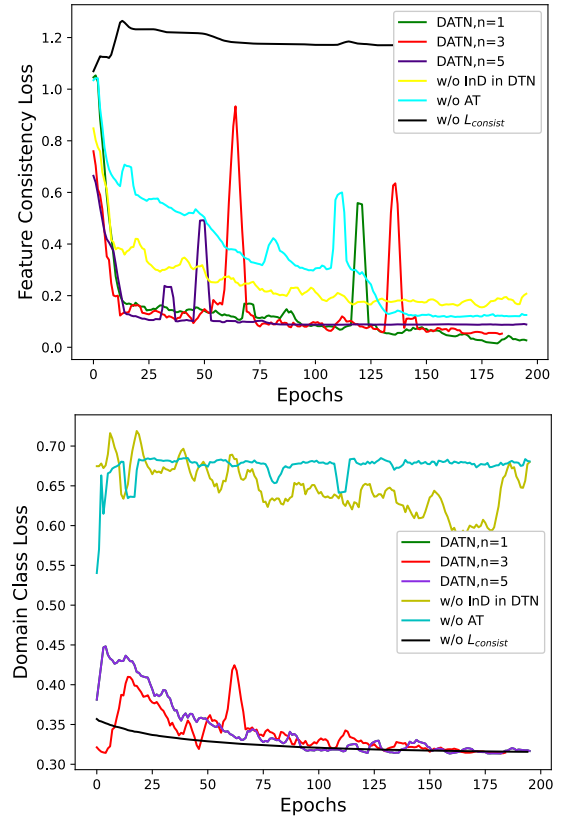


Fig. 6. Training curve of the feature consistency loss $\mathcal{L}_{consist}$ and the domain class loss $\mathcal{L}_{class}$ when training DATN with different ablation schedules on the LOL dataset.

TABLE VII

ANALYSIS OF THE IMPACT OF DATN ON LOL DATASET IN TERMS OF PSNR/SSIM. THE BEST RESULTS ARE HIGHLIGHTED IN **BOLD**, AND THE SECOND BEST RESULTS ARE IN UNDERLINE

Schedule	PSNR	SSIM
IFCN $\rightarrow \times 1 \rightarrow$ Raw-InDNet	8.48	0.277
IFCN $\rightarrow \times 5 \rightarrow$ Raw-InDNet	12.70	0.573
IFCN $\rightarrow \times 10 \rightarrow$ Raw-InDNet	16.74	0.694
IFCN $\rightarrow \times 15 \rightarrow$ Raw-InDNet	19.24	0.736
IFCN $\rightarrow \times 20 \rightarrow$ Raw-InDNet	19.66	0.746
IFCN $\rightarrow \times 25 \rightarrow$ Raw-InDNet	17.50	0.741
IFCN $\rightarrow \times 30 \rightarrow$ Raw-InDNet	15.79	0.728
IFCN $\rightarrow$ DATN $\rightarrow$ Raw-InDNet	<u>20.11</u>	<u>0.755</u>
IFCN $\rightarrow$ DATN-MGC $\rightarrow$ Raw-InDNet	20.42	0.761

decouple the entire framework, keep IFCN and Raw-InDNet fixed, remove DATN, and directly send the output of IFCN to Raw-InDNet for enhancement. We refer to this schedule as IFCN $\rightarrow \times 1 \rightarrow$ Raw-InDNet, as in the first row of Table VII. This obviously achieves poor results, because even if the format differences are eliminated, there are still considerable distribution differences between domains.

Then, in the next few rows, we perform numerical multiplication on the IFCN-converted Raw data and then send it to the Raw-InDNet. Here in the second row, IFCN $\rightarrow \times 5 \rightarrow$ Raw-InDNet means the output data of IFCN is magnified five times before being sent to Raw-InDNet. Other rows can be interpreted similarly. As the magnification continues to increase, we can observe quite good results at a certain magnification, such as the fifth row with a magnification of 20. These results

TABLE VIII

ABLATION STUDY RESULTS OF THE ORDER n OF DTN IN TERMS OF PSNR/SSIM. THE BEST RESULTS ARE HIGHLIGHTED IN **BOLD**, AND THE SECOND BEST RESULTS ARE IN UNDERLINE. NOTE THAT THE INFERENCE TIME IS MEASURED ON 600×400 IMAGES OF THE LOL DATASET

Order n of DTN	PSNR	SSIM	Time
1	18.39	0.741	0.023s
2	19.44	0.748	0.025s
3	20.11	<u>0.755</u>	0.028s
4	<u>19.83</u>	0.758	0.031s
5	18.76	0.754	0.033s

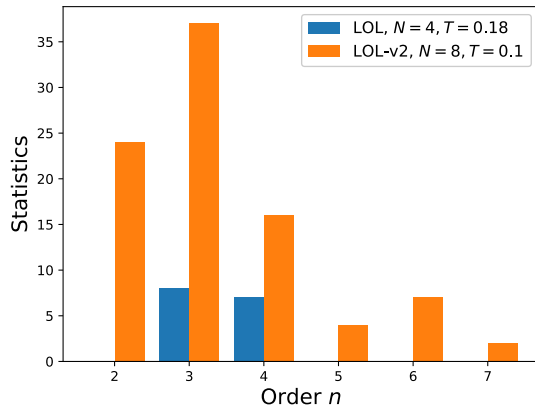


Fig. 7. Statistics of the actual number of iterations n of DTN when using MGC on LOL and LOL-v2 test sets.

illustrate that within our proposed framework, using IFCN for format conversion and Raw-InDNet for enhancement, even with the simplest numerical amplification, we can obtain competitive cross-domain LLIE results. However, how can we obtain accurate magnification, and is numerical magnification the optimal solution? In this article, we design DATN to try to answer this question. As we can see from Table VII, DATN and DATN+MGC achieve the best results. We use unsupervised DATN to make up for the across-domain overall brightness difference and other potential distribution differences.

3) Impact of MGC and the Order of DTN: In this ablation study, we first report the impact of the order of DTN, i.e., the iteration number n in (11), on cross-domain LLIE performance and the training process. Note that when we report the results of CUDA without MGC, the same fixed n value is used in the test phase as in the training phase. It can be seen from Table VIII that as the order n increases, PSNR and SSIM results reaches the peak value when $n = 3$ and $n = 4$ then fall back, while the inference time keeps increasing accordingly. Also, no significant correlation was observed in Fig. 6 between the order and the training convergence speed. Therefore, combining the reported performance and the inference time, we set the order n to 3 for unsupervised training and inference of the CUDA framework in this article.

As for the role of MGC, since it involves hyperparameters that need to be set based on experience, that is, the maximum iteration number N in inference and the iteration termination mean threshold T , we do not report it as the formal result of our CUDA framework but only use it as a supplement,

showing more possibilities of the DATN method. Fig. 7 shows the statistics of the actual number of iterations n on LOL and LOL-v2 test sets when appropriate termination threshold N and T is selected. It can be seen that most of the iterations are around the n value used for training, but there are also some images that require more or fewer iterations. Combined with the excellent results of CUDA-MGC reported in Table III, it can be concluded that DATN provides a good foundation and more targeted gain adjustment can achieve better performance improvements.

VII. LIMITATIONS AND ADVANTAGES DISCUSSION

Although our proposed InDNet and CUDA framework have achieved satisfactory results in handling both supervised and unsupervised LLIE tasks, there are still some limitations in terms of model complexity, parameter tuning, along with relatively low LPIPS values for unsupervised RGB results that merit further investigation in future research.

Compared to other supervised LLIE methods, the advantages of our proposed InDNet lie in the plug-and-play nature of the InD module, which can be easily integrated into existing network frameworks with minimal parameter costs to achieve significant performance improvements. In addition, the plug-and-play feature of the InD module allows it to adapt to various network structures. For example, we directly introduce it into the DATN network of the CUDA framework for integrating global information for domain adaptation. Therefore, we believe that the InD module has the potential to be applied to other network structures or fields. The limitations of the InD module mainly stem from the design of the module, where multiple matrix multiplications are used to fully integrate information. This may lead to optimization issues when too many InD modules are inserted in practical use, which we briefly discuss in Section VI-A2. In the future, more efficient designs for information integration computation may resolve this issue.

The advantages of our proposed CUDA framework are attributed to leveraging well-trained Raw domain LLIE networks. Through domain adversarial transfer, it requires only minimal batches of unsupervised training on the target datasets to achieve competitive unsupervised RGB format LLIE results. The high-quality Raw domain dataset pretrained network, InDNet, serves as the predictor within the CUDA framework, resulting in superior noise suppression on the target dataset compared to typical unsupervised LLIE models, as depicted in Fig. 5. Well-designed DATN ensures that the illumination enhancement effect of the CUDA framework is superior to other comparative methods, as also evident from the comparisons in Fig. 5.

The limitations of our proposed CUDA framework mainly stem from the three-stage design. The CUDA framework consists of IFCN, DATN, and Raw-InDNet as the predictor. Since we fixed IFCN and the predictor during training, the training process is straightforward. However, during testing, input images need to undergo the three-stage network, resulting in higher computational costs and longer testing times compared to one-stage models. In addition, the performance of each stage in the three-stage process can affect the overall

performance of the framework. For example, the pretrained Raw-InDNet, which serves as the predictor, contains priors from the high-resolution extreme low-light (high noise level) SID dataset. This leads to occasional over-denoising issues when testing on images with low noise levels and resolutions, as reflected in the relatively lower LPIPS results in Table III. Due to the inherent differences between Raw and RGB formats, IFCN still serves a necessary purpose. Therefore, our next step could involve integrating DATN and the predictor to enhance the adaptability of the source domain model to target domain data, while reducing the number of inference stages. This approach may have the potential to further improve the performance ceiling of the framework.

VIII. CONCLUSION

In this work, we contribute to the LLIE task in two dimensions. First, we present an InD module that gathers pixel-intensive global information efficiently to tackle the challenges of detail and color reconstruction in supervised Raw and RGB format LLIE tasks. On top of this, we build a novel RGB-to-Raw CUDA framework with a DATN to bridge the domain gap between Raw and RGB format LLIE tasks. Resulting in successfully tackling the unsupervised RGB format LLIE task by fully leveraging the Raw priors inherent in the pretrained Raw domain LLIE network. Comprehensive experimental results demonstrate that LLIE networks equipped with InD modules acquire great performance improvement when trained with paired images and our CUDA framework also achieves state-of-the-art unsupervised results on RGB datasets. Moreover, we analyzed the limitations and advantages of the proposed approaches, as well as its potential for further development. We hope that our method and framework inspire new investigations along these research directions.

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Pengliang Tang received the bachelor's degree in radio and television engineering from the Nanjing University of Posts and Telecommunications, Nanjing, China, in 2016. He is currently pursuing the Ph.D. degree with the School of Artificial Intelligence, Beijing University of Posts and Telecommunications, Beijing, China.

His current research interests include digital image processing and low-light image enhancement.

Jiangbo Pei received the bachelor's degree in communication engineering from the Beijing University of Posts and Telecommunications, Beijing, China, in 2020, where he is pursuing the Ph.D. degree.

He is also affiliated with the National Institute of Health Data Science, Peking University, Beijing. His research interests include computer vision, unsupervised learning, and transfer learning.

Jianan Han received the bachelor's degree in communication engineering from Northeastern University, Shenyang, China, in 2016. He is currently pursuing the Ph.D. degree with the School of Artificial Intelligence, Beijing University of Posts and Telecommunications, Beijing, China.

Concurrently, he is interning with the National Institute of Health Care Big Data, Peking University, Beijing. His current research focuses on medical vision analysis and biomedical multimodal data analysis.

Aidong Men received the B.S., M.S., and Ph.D. degrees from the Department of Radio Engineering, Beijing University of Posts and Telecommunications, Beijing, China, in 1987, 1990, and 1994, respectively.

He is currently serving as a Professor with the Intelligent Media Computing Teaching and Research Center, School of Artificial Intelligence, Beijing University of Posts and Telecommunications. His research areas encompass deep learning, digital broadcasting, multimedia communication, and related fields.